Project Atlas

Master Flightline Clock

v1.6.0 ACTIVE DEVELOPMENT Last Updated: July 2026 | Python 3.12 | PySide6 6.11.1

Overview

Project Atlas is the development codename for the ATLAS Master Flightline Clock — a specialised timing and display system designed to manage competitive F3K model glider race heats for the Southern Soaring League.

ATLAS provides a deterministic, operator-controlled timing appliance that synchronises with the host computer clock and drives a highly visible public display for pilots and event staff.

Purpose

ATLAS manages the following race-day operations:

- Real-time clock display
- Heat scheduling and persistence
- Pilot insertion window timing
- Heat countdown timing
- Public-facing race status display
- Operator-controlled schedule maintenance
- Persistent competition settings and heat data

The system is intended to behave as a dedicated field appliance rather than a general-purpose desktop application.

Current Functional Status

ATLAS is currently a working Python desktop application with:

- A PySide6 operator console (three-tab: LIVE / SCHEDULE / SETTINGS)
- SQLite-backed persistence for heats and settings
- Automatic heat activation logic
- Phase-based countdown display logic

- A software LED simulator display backend
- A full-resolution LCD monitor display backend
- Settings persistence for display mode, display size, brightness, and auto-boot preference

Architecture Overview

Location	Module	Role
`src/main.py`	main	Application entry point and runtime wiring
`src/core/engine.py`	engine	Timing engine and state logic
`src/core/database.py`	database	SQLite persistence and settings
`src/core/models.py`	models	Data models (Pydantic)
`src/core/config.py`	config	Legacy JSON config manager (not active)
`src/core/timers.py`	timers	Legacy timer module (not active)
`src/gui/windows.py`	windows	Operator console UI
`src/gui/led_simulator.py`	led_simulator	LED-style software display backend
`src/gui/lcd_display.py`	lcd_display	Full-resolution monitor display backend
`src/gui/fonts.py`	fonts	Bitmap font definitions
`src/gui/themes.py`	themes	UI theme and CSS styling

Technology Stack

- Python 3.12
- PySide6 6.11.1
- SQLite
- Pydantic 2.x
- Git / GitHub

Running ATLAS

From the project root:

```
source .venv/bin/activate
export PYTHONPATH=$PYTHONPATH:$(pwd)/src
python src/main.py
```

Alternatively, use the helper launcher:

```
./run_atlas.sh
```

Display Backends

ATLAS supports two interchangeable public display renderers, selectable from the Settings tab:

- **LED Simulator** — software simulation of a pixel LED board (128x64 or 64x64)
- **LCD Monitor Display** — full-resolution anti-aliased display for standard monitors and TVs

Settings Storage

> The current live source of truth for settings is the SQLite database, not `atlas_config.json`. The JSON config file and `config.py` remain in the repository for historical continuity but are not in the active runtime path.

Legacy Modules

The following files are present but not in the active runtime path:

- `src/core/config.py`
- `atlas_config.json`
- `src/core/timers.py`

Development Status

ATLAS is beyond the initial foundations stage and is now a functioning application under active refinement. Current focus areas include:

- Deployment hardening and cross-platform deployment process
- Documentation refresh and formalisation
- Appliance-mode behaviour

- Future physical display hardware integration (P5/P10 LED panels)

Design Priorities

- Reliability and deterministic behaviour
- Operational clarity for field use
- Maintainability
- Appliance-style deployment
- Future hardware extensibility